

Young women taking SSRI antidepressants may be at increased risk of suicidal behavior

Stockholm, 4 June 2026. Initiating a selective serotonin reuptake inhibitor (SSRI) within 28 days of a first depression diagnosis was estimated to raise 12-week suicidal-behaviour risk by +0.97 percentage points (95% CI 0.78, 1.17) among Swedish young people aged 6 to 24, with the increase roughly 2.2× as large in females as in males, according to a new master's thesis from Karolinska Institutet.

The analysis followed 88,427 individuals with a first depression diagnosis between 2006 and 2019, drawing on Sweden's national health, prescription, and population registers. It used target-trial emulation – a method that derives trial-like causal estimates from observational data – together with iterative causal forests, a recently developed machine-learning technique that searches for patient subgroups who respond differently to treatment.

After adjusting for differences in patient characteristics between those who did and did not start treatment, the algorithm discovered a meaningful distinction between male and female patients. The estimated 12-week absolute risk increase was +0.54 percentage points in males (95% CI 0.26, 0.82) and +1.17 percentage points in females (0.91, 1.43). Searches across additional diagnostic codes and prescription patterns produced no further reproducible subgroups.

“The risk increase in the first 12 weeks is small overall, but it is significantly larger in female patients than in male patients,” said Ville Sebastian Olsson, the thesis author. “Sex should be taken into consideration when starting one of these drugs, not just age.”

The FDA first issued a black-box warning on SSRIs in 2004 after a meta-analysis of paediatric trials found roughly a two-fold higher rate of suicidal ideation or attempts in treated children and adolescents. That warning stratifies on age, not on sex. The new thesis adds sex as a candidate additional axis of caution in a national real-world cohort more than two decades after the original trial-based finding.

The thesis emphasizes that the contrast is between starting an SSRI and not starting one, not between SSRIs and alternative treatments such as psychotherapy. It is also noted that the absolute increase is small (roughly one additional event per 100 patients treated over 12 weeks). Replication in independent registries, longer follow-up, and an active-comparator analysis are listed as the natural next steps.

Contact:

Ville Sebastian Olsson, thesis author

Department of Medical Epidemiology and Biostatistics, Karolinska Institutet

Email: ville.sebastian.olsson@ki.se

Statement on the use of generative AI

I used OpenAI ChatGPT to assist in providing feedback on the written text and helping me refine it.